1.

a. Four continuous variables p1, p2, p3, and p4 give the output of each plant in MW. The objective minimizes total cost, $0.02p1^2 + 8p1 + 0.04p2^2 + 12p2 + 0.05p3^2 + 15p3 + 0.025p4^2 + 5p4$, subject to $p1 + p2 + p3 + p4 = 600$ for demand, the capacities $p1 \le 350$, $p2 \le 200$, $p3 \le 200$, and $p4 \le 150$, and nonnegativity $p1, p2, p3, p4 \ge 0$.
b. The optimal plan runs plant 1 at 300 MW, plant 2 at 100 MW, plant 3 at 50 MW, and plant 4 at its 150 MW maximum, for a total cost of $7,987.50 per hour.
c. Running every plant at 150 MW costs 1,650 + 2,700 + 3,375 + 1,312.50 = $9,037.50 per hour, so the optimal plan saves $1,050 per hour.
d. The marginal costs are $2(0.02)(300) + 8 = 20$ for plant 1, $2(0.04)(100) + 12 = 20$ for plant 2, $2(0.05)(50) + 15 = 20$ for plant 3, and $2(0.025)(150) + 5 = 12.5$ for plant 4. Plants 1 to 3 all cost $20 for one more MW, so moving power between them cannot lower the cost. Plant 4 costs only $12.50 for one more MW, which is cheaper than any other plant, so it would keep taking output and only stops because it reaches its 150 MW capacity.

2.

a. $\nabla f(x, y) = (3x^2 - 12, 3y^2 - 3)$ and $H(x, y) = [6x\ 0;\ 0\ 6y]$.
b. Setting $\nabla f = 0$ gives $3x^2 = 12$, so $x = \pm 2$, and $3y^2 = 3$, so $y = \pm 1$. The stationary points are $(2, 1)$, $(2, -1)$, $(-2, 1)$, and $(-2, -1)$.
c. H is diagonal, so its eigenvalues are 6x and 6y.
$(2, 1)$: $H = [12\ 0;\ 0\ 6]$, both positive, so it is a local minimum with $f = -18$.
$(-2, -1)$: $H = [-12\ 0;\ 0\ -6]$, both negative, so it is a local maximum with $f = 18$.
$(2, -1)$: $H = [12\ 0;\ 0\ -6]$, mixed signs, so it is a saddle point with $f = -14$.
$(-2, 1)$: $H = [-12\ 0;\ 0\ 6]$, mixed signs, so it is a saddle point with $f = 14$.
d. f is neither convex nor concave, since H is positive definite only when $x > 0$ and $y > 0$, negative definite only when $x < 0$ and $y < 0$, and indefinite otherwise. The local minimum at $(2, 1)$ is not a global minimum because the x^3 term makes f unbounded below as $x \to -\infty$, for example $f(-10, 0) = -1000 + 120 = -880 < -18$.

3.

a. $\partial f/\partial a = -2[(1 - a) + (3 - a - b) + (4 - a - 2b) + (7 - a - 3b) + (8 - a - 4b)] = -2(23 - 5a - 10b) = 10a + 20b - 46$.
$\partial f/\partial b = -2[(3 - a - b) + 2(4 - a - 2b) + 3(7 - a - 3b) + 4(8 - a - 4b)] = -2(64 - 10a - 30b) = 20a + 60b - 128$.
b. At $(0, 0)$, $f = 139$ and $\nabla f = (-46, -128)$.

Iteration 1: (a, b) = (0, 0) − 0.01(−46, −128) = (0.46, 1.28). f = 15.986, ∇f = (−15.8, −42), ||∇f|| = 44.87.
Iteration 2: (a, b) = (0.46, 1.28) − 0.01(−15.8, −42) = (0.618, 1.70). f = 2.594, ∇f = (−5.82, −13.64), ||∇f|| = 14.83.

c. With α = 0.01, gradient descent met the stopping rule after 154 iterations, at a = 0.9969 and b = 1.8011, with f = 0.8000. Gurobi gives a = 1, b = 1.8, and f = 0.8, so the fitted line is y = 1 + 1.8x.
d. Halving the learning rate to 0.005 roughly doubles the iterations, and doubling it to 0.02 roughly halves them, since each step moves in proportion to α. At 0.035 each step overshoots the minimum and lands farther away than before, so f grows to about 10^{259} and the run hits the 1000 iteration cap without converging. The Hessian [10 20; 20 60] has a largest eigenvalue of about 67, so gradient descent only converges when α < 2/67 ≈ 0.03.

Learning rate	Iterations	Result
0.005	309	Converged, f = 0.8
0.01	154	Converged, f = 0.8
0.02	76	Converged, f = 0.8
0.035	1000	Diverged